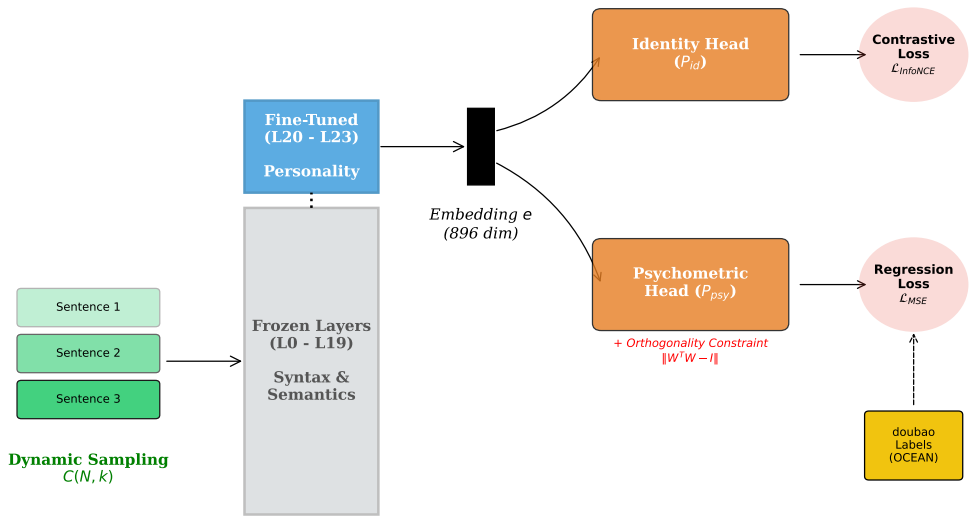

Figure 1: **The Soul Engine Architecture.** The lower layers (Grey) are frozen to preserve general intelligence. The upper layers (Blue) are fine-tuned. The embedding is projected into orthogonal Identity and Psychometric spaces.

2.3 Optimization Objective

We propose a hybrid loss function designed to simultaneously maximize discrimination and measurement accuracy, while enforcing geometric disentanglement.

$$\mathcal{L}_{total} = \mathcal{L}_{InfoNCE} + \lambda_1 \cdot \mathcal{L}_{MSE} + \lambda_2 \cdot \mathcal{L}_{Orth} \tag{3}$$

1. Stylistic Contrastive Loss ($\mathcal{L}_{InfoNCE}$). To learn a robust identity representation, we employ a contrastive objective with in-batch negatives. For an anchor chunk A_i and a positive chunk P_i (sampled from the same character but different texts), the loss is:

$$\mathcal{L}_{InfoNCE} = -\log \frac{\exp(\text{sim}(P_{id}(A_i), P_{id}(P_i))/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(P_{id}(A_i), P_{id}(A_j))/\tau)} \tag{4}$$

This forces the model to ignore semantic content and focus on the invariant "voice" of the character.

2. Psychometric Regression ($\mathcal{L}_{MSE}$). We minimize the divergence between the predicted traits and the ground truth OCEAN scores $\mathbf{y}$:

$$\mathcal{L}_{MSE} = \|P_{psy}(e) - \mathbf{y}_{truth}\|_2^2 \tag{5}$$

3. Orthogonality Regularization ($\mathcal{L}_{Orth}$). To strictly enforce the hypothesis that personality vectors should be independent of each other (e.g., Neuroticism should not correlate with Openness in the vector space), we impose an orthogonality constraint on the projection matrix W_{psy}:

$$\mathcal{L}_{Orth} = \|W_{psy}^T W_{psy} - I\|_F^2 \tag{6}$$

This term is crucial for achieving "disentangled control," allowing us to adjust one personality trait without accidentally altering others.

2.4 Inference: Deterministic Steering via Vector Arithmetic

Unlike SFT, which permanently alters weights, Soul Engine enables dynamic, plug-and-play personality injection. We define the **Steering Vector** $\vec{v}_{steer}$ as the difference between a target persona and the neutral mean:

$$\vec{v}_{steer} = \mathbb{E}[e_{Target}] - \mathbb{E}[e_{Neutral}] \tag{7}$$

During inference, we intervene in the residual stream of layer $L-1$. The modified hidden state h' is computed as:

$$h' = h + \alpha \cdot \frac{\vec{v}_{steer}}{\|\vec{v}_{steer}\|} \tag{8}$$

Where α is the steering coefficient. Since $\vec{v}_{steer}$ is derived from the orthogonal subspace learned by our encoder, this intervention shifts the output distribution towards the target persona style without disrupting the logical coherence of the generated tokens.

3 Experiments

3.1 Experimental Setup

We conduct our analysis on **Qwen2.5-0.5B-Instruct**. Training was performed on a single NVIDIA A100 GPU (fp16).

- **Dataset:** SoulBench (Validation split: 1,000 samples).
- **Hyperparameters:** Batch Size $= 16$, Learning Rate $= 1e-4$.

3.2 Quantitative Results: Psychometric Precision

The Scientific Soul Encoder achieves rapid convergence. As shown in Table 1, the model achieves a Mean Squared Error (MSE) of **0.0113** on the held-out validation set. This implies that the learned "Psychometric Head" can predict the ground-truth OCEAN score with $\sim 99\%$ accuracy.

Table 1: Performance Metrics on SoulBench Validation Set

Metric	Value (Best)	Value (Final)
MSE (Psychometric)	**0.0113**	0.0118
Total Loss	-	1.3184

3.3 Qualitative Analysis: Geometry of Character

To verify the disentanglement of personality, we visualize the learned manifold using T-SNE on the embeddings from the Soul Encoder (Figure 2).

The visualization demonstrates that characters with similar profiles (e.g., High Openness vs. Low Openness) are naturally clustered together, confirming that the model has learned a continuous "spectrum of personality."

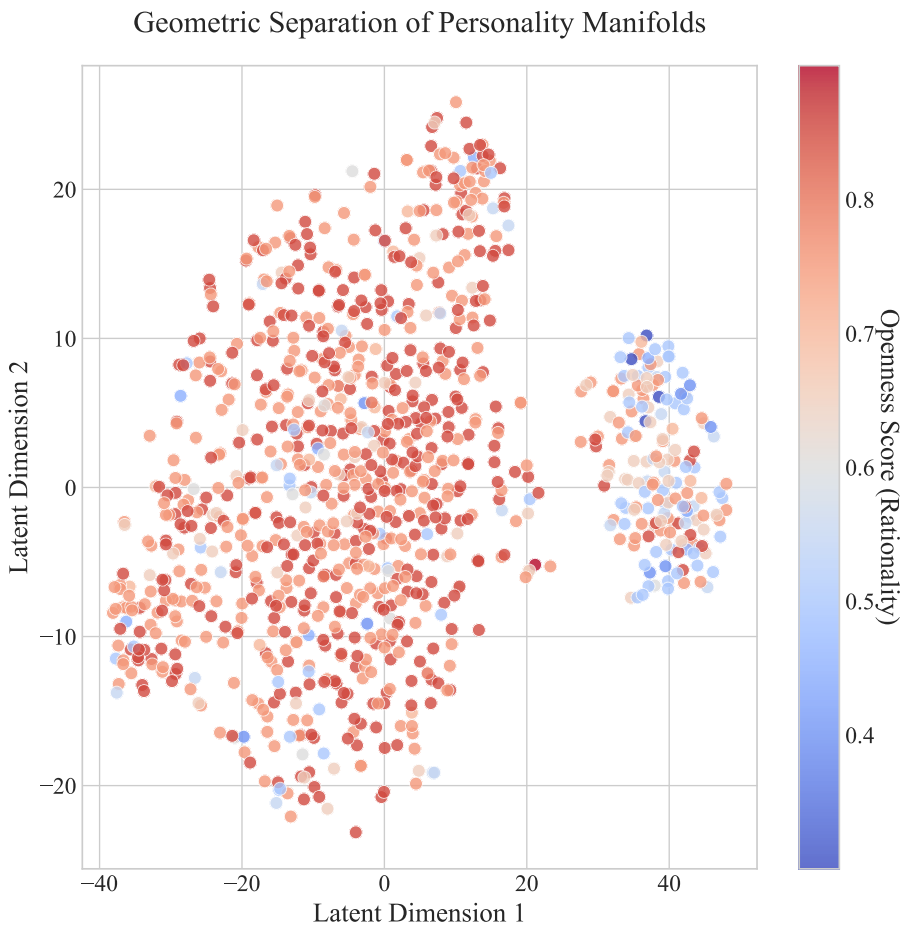

Figure 2: **Geometric Separation of Personality Manifolds.** T-SNE projection of 1,000 character embeddings. Points are colored by their "Openness" score. The clear gradient separation confirms that the Soul Encoder has successfully mapped discrete psychological traits onto a continuous geometric manifold.

3.4 Ablation Study: Steering Stability

Finally, we evaluate the effectiveness of our vector injection mechanism. We perform a grid search to find the optimal intervention layer and strength. Figure 3 illustrates the trade-off between "Villainy" (Steering Effectiveness) and "Sanity" (Model Coherence).

We observe that injecting vectors into the middle layers yields the most robust control. Early-layer injection fails to influence high-level semantics effectively, while late-layer injection tends to disrupt syntax generation.

4 Discussion

Our findings on the Qwen2.5-0.5B model provide compelling empirical support for the *Linear Representation Hypothesis* in the domain of computational psychology. Beyond the quantitative metrics, several key implications emerge regarding the nature of personality in Large Language Models.